

1. Prerequisites

A knowledge of the specifications for P1, P2, P3, P4 and M1 and M2, and their prerequisites and associated formulae, is assumed and may be tested.

2. Examination

- First assessment: June 2020.
- The assessment is 1 hour and 30 minutes.
- The assessment is out of 75 marks.
- Students must answer all questions.
- Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*.
- The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.

3. Notation and formulae

Students will be expected to understand the symbols outlined in *Appendix 7: Notation*.

Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

$$\text{The tension in an elastic string} = \frac{\lambda x}{l}$$

$$\text{The energy stored in an elastic string} = \frac{\lambda x^2}{2l}$$

For SHM:

$$\ddot{x} = -\omega^2 x,$$

$$x = a \cos \omega t \text{ or } x = a \sin \omega t,$$

$$v^2 = \omega^2(a^2 - x^2),$$

$$T = \frac{2\pi}{\omega}$$

M3.3 Unit content

What students need to learn:		Guidance
1. Further kinematics		
1.1	Kinematics of a particle moving in a straight line when the acceleration is a function of the displacement (x), or time (t).	The setting up and solution of equations where $\frac{dv}{dt} = f(t), v \frac{dv}{dx} = f(x), \frac{dx}{dt} = f(x) \text{ or } \frac{dx}{dt} = f(t)$ will be consistent with the level of calculus required in units P1, P2, P3 and P4.